In this type of breadboard, two, long, power busses on both sides of the breadboard are useful for providing a + and – power supply rail that runs along the length of the board. On either side of a channel that runs down the center of the board are two rows of holes where each row of five holes are connected to each other. This center divider keeps these two sides separate and provides a place for integrated circuits that typically have two rows of pins to be plugged into the board and still be connected to other devices without inadvertently connecting the two sides of the IC together. Keep in mind that any pin inserted into one of these adjacent five positions will be connected to one another inside of the breadboard.

Now, to get a better sense for how can we use the breadboard to prototype a basic circuit, let's try one out. Figure 12-14 takes our basic lamp circuit and changes out a few of the components.

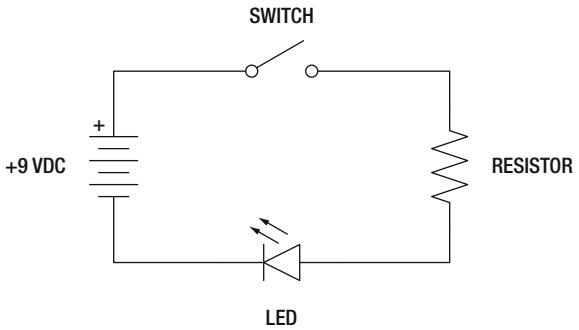


Figure 12-14. *Basic LED circuit schematic*

In this circuit, we have replaced the incandescent lamp with a light-emitting diode and the necessary current limiting resistor, but otherwise the circuit will work just the same as before. Now let's look at Figure 12-15 to see what it looks like on the breadboard.

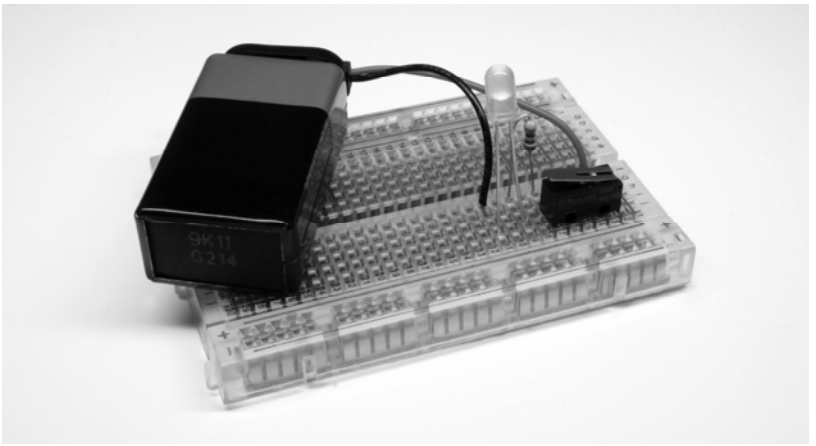


Figure 12-15. LED circuit in a breadboard